A Literature-Implied Consistent Counterexample to the Corson Property (M) Equivalence

Math discovery status packet for arXiv:1912.08564

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Status and source question

Status: literature-implied answer (consistent negative under Jensen's diamond).

In Remark 3.6 on PDF page 9 of [1], Correa asks whether, for every Corson compactum K,

$$C(K) \text{ has the } c_0\text{-extension property} \quad \Longleftrightarrow \quad K \text{ has Property (M).}$$

The same remark asks specifically whether $C(K)$ has the extension property for Kunen's Corson compactum without Property (M).

The universal equivalence is consistently false. The conclusion follows by combining Correa's Theorem 2.5 with Plebanek's later Theorem 2.9; the latter paper does not state the resulting c_0-extension corollary, so the identification is literature-implied rather than an explicit literature answer.

Identification

Plebanek proves under Jensen's diamond that there is a Corson compact space K such that $P(K)$, the compact space of probability measures on K, is $\aleph_0$-monolithic, while K supports a probability measure μ of Maharam type ω_1 [2, Theorem 2.9, PDF p. 4]. The introduction of the same paper records the equivalence

$$P(K) \text{ is } \aleph_0\text{-monolithic} \quad \Longleftrightarrow \quad (B_{C(K)^*}, w^*) \text{ is } \aleph_0\text{-monolithic.}$$

Correa's Theorem 2.5 therefore applies and shows that $C(K)$ has the 2-c_0-extension property [1, Theorem 2.5, PDF p. 6].

It remains only to check that K fails Property (M). If $\operatorname{supp}\mu$ were separable, then it would be metrizable: Corson compacta are monolithic, and the support is a closed subspace of K. But every finite measure on a compact metrizable space has countable Maharam type, because its Borel sigma-algebra is countably generated modulo completion (equivalently, its L_1-space is separable). This contradicts the fact that μ has type ω_1. Hence $\operatorname{supp}\mu$ is nonseparable, so K does not have Property (M).

Consequently, under diamond,

$$C(K) \text{ has the 2-}c_0\text{-extension property} \qquad \text{but} \qquad K \text{ fails Property (M).}$$

Scope and search status

This consistent counterexample shows that the universal Corson equivalence cannot be proved in ZFC, assuming the consistency of ZFC with diamond. It does not produce a ZFC example. Plebanek explicitly notes that it was not known whether the construction follows from CH alone. It also does not settle the case of Kunen's particular compactum.

Nor does the example answer Correa's separate question whether the c_0-extension property of an arbitrary Banach space forces its dual ball to be weak-star monolithic: the dual ball in Plebanek's example is monolithic, so the hypothesis of Correa's sufficient theorem is satisfied.

The bounded status check searched the run's four lightweight indexes for both arXiv ids and the core terms, and searched the web for the exact source title, the converse question, the Corson equivalence, Property (M), and Kunen. The decisive evidence is the locally ingested arXiv source and PDF of Plebanek's Theorem 2.9 together with its introductory dual-ball equivalence.

Human-review recommendation. Check the dual-ball equivalence, the application of Correa's Theorem 2.5, and the one-paragraph Maharam-type argument. Classify the result as a consistent negative answer, while leaving the ZFC, CH-only, Kunen-specific, and general Banach-space converse questions open.

References

[1] C. Correa, *On the c_0-extension property*, Studia Mathematica 256 (2021), 345–359, arXiv:1912.08564.

[2] G. Plebanek, *Monolithic spaces of measures*, arXiv:1912.13297.